

DATABASE MANAGEMENT SYSTEM LAB

Assignment No. 10 (-Name : Ridafatema Patel, Rollno. 63)

iot@iot-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~\$ mongosh

```
test> use RIDA
switched to db RIDA
RIDA> db.Book.find();
```

Q.1 Find the number of books published by Avhad Mam.

```
RIDA> db.Book.aggregate({$match : {By : "Kalyani Mam"}},{ $count : "TotalBooks"})
[ { TotalBooks: 1 } ]
RIDA> db.Book.aggregate({$match : {By : "Avhad Mam"}},{ $count : "TotalBooks"})
[ { TotalBooks: 2 } ]
```

Q.2 Find books which have minimum likes and maximum likes published by Avhad Mam.

```
RIDA> db.Book.aggregate([{$match : {By : "Avhad Mam"}},{ $group : { _id : "$By", MinLikes :
{$min : "$Likes"},MaxLikes : {$max : "$Likes"}}}])
[ { _id: 'Avhad Mam', MinLikes: '50', MaxLikes: '50' } ]
```

Q.3 Find the average number of likes of the books published by Avhad Mam.

```
RIDA> db.Book.aggregate([{$match : {By : "Avhad Mam"}},{ $group : { _id : "$By", AvgLikes :
{ $avg : "$Likes"}}}])
[ { _id: 'Avhad Mam', AvgLikes: null } ]
```

note : This is a **string**, not a number → \$avg cannot calculate → result = null

Q.4 Find the first and last book published by Avhad Mam.

```
RIDA> db.Book.aggregate([{$match : {By : "Avhad Mam"}},{ $group : { _id : "$By", Last : {$last :
"$Likes"},First : {$first : "$Likes"}}}])
[ { _id: 'Avhad Mam', Last: '50', First: '50' } ]
RIDA> db.Book.aggregate([{$match : {By : "Avhad Mam"}},{ $group : { _id : "$By", Last : {$last :
"$Title"},First : {$first : "$Title"}}}])
[ { _id: 'Avhad Mam', Last: 'DBMS', First: 'DBMS' } ]
RIDA> db.Book.aggregate([{$match : {By : "Kalyani Mam"}},{ $group : { _id : "$By", Last : {$last :
"$Title"},First : {$first : "$Title"}}}])
[ { _id: 'Kalyani Mam', Last: 'OOP', First: 'OOP' } ]
```

Q.5 Create an index on author name.

```
RIDA> db.Book.createIndex({ By: 1 })
By_1
RIDA> db.Book.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { By: 1 }, name: 'By_1' }
]
RIDA> db.Book.find({ By: "Avhad Mam" }).explain("executionStats")
```